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Historical changes and role of river discharge in water quality: 30 years of environmental monitoring in Uruguay River

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Abstract

Background Land-use intensification has enhanced impacts on water quality of South American rivers. A long-term database was compiled from public monitoring data to evaluate the temporal variation in water quality in the mid-lower Uruguay River from 1987 to 2019. Forty-two parameters (physicochemical, nutrients, metals, microbiological, pesticides, persistent, bioaccumulative and toxic substances and hydrological) in three sites across 440 km of the river were selected. Bayesian Generalized Linear Models were applied to understand the temporal and discharge-related drivers of each water quality parameter. Finally, a water quality index (CCME-WQI) was calculated to evaluate temporal trends in water quality.

Results Most of the parameters (70%) exceeded the regulation values at least once during the 30-year period. Pesticides like aldrin, DDT, dieldrin, heptachlor and lindane, and nutrients (Kjeldahl nitrogen and total nitrogen) showed higher average values in the 1990s. Cadmium, chromium, copper, lead and mercury showed the highest averages in the 2000s, while iron concentration was the highest in the 2010s. The response of water quality parameters to time and discharge varied across sites, with most of the effects observed upstream Salto Grande reservoir. In all three sites, calcium, BOD₅, and magnesium decreased over time, while iron and nitrite increased. The 30-year CCME-WQI revealed an improvement in water quality (from Poor to Good) over time, concomitant with the ban of persistent organic contaminants towards the 2000s (e.g., dieldrin and lindane). However, a CCME-WQI calculated for a recent period (2018–2019) from a monitoring program with 181 parameters and a different parameter selection setup, showed barely Marginal and Good values, highlighting the sensitivity of the index to the parameters included.

Conclusions During the 30-year period, the water quality of the mid-lower Uruguay River was never ranked as excellent, although improvements were observed in specific parameters (i.e. banned pesticides) increasing overall water quality. However, several water quality parameters (i.e. metals and nutrients) remain above the regulations recommended levels with increasing concentrations over time. The results provide valuable information to support environmental management and highlight the increasingly complex challenges of water quality dynamics in large river systems.

Keywords Water quality trend, Hydrological regime, Bayesian modeling, Pampas biome, Water quality index

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Background

The structure and functioning of freshwater ecosystems are affected by global drivers such as climate change, land-use intensification, habitat degradation, urbanization, and damming (Allan 2004; Moss 2009; Sabater et al. 2019; Luo et al. 2020; Torremorell et al. 2021; Brauns et al. 2022). Land-use intensification has led to

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an increase in nutrient loading and agricultural pollutants (e.g., pesticides, herbicides) in aquatic systems (Smith and Schindler 2009; Paerl 2017; Sharma et al. 2019; Wurtsbaugh et al. 2019) and a reduction in riparian buffer zones (Aguiar et al. 2015; Mello et al. 2018). Urban wastewater discharges introduce organic matter, nutrients, and microbiological contaminants (e.g., fecal coliforms) into receiving water bodies (Jones and Stanley 2016), as well as emerging contaminants such as pharmaceutical contaminants (e.g., antibiotics, analgesics, and hormones) (Hughes et al. 2013; Cantwell et al. 2018), drugs, and personal care products, which may have physiological impacts on aquatic biota (e.g., endocrine disruption) (Petrie et al. 2015; aus der Beek et al. 2016; Ebele et al. 2017; Liess et al. 2021) and many of them without reference values in the regulations (Torremorell et al. 2021). Eutrophication is one of the main consequences of both point and diffuse organic and inorganic pollution (Smith and Schindler 2009). Additionally, industrial activities can increase the concentration of metals and persistent pollutants (e.g., adsorbable organic halogen compounds, dioxins, and phenols) (Allan and Castillo 2007). In general, all these chemical compounds occur as a complex mixture originating from both current and historical watershed activities (Harding et al. 1998; Inamdar et al. 2021) with potential synergistic effects (Petrie et al. 2015), raising major challenges in monitoring and managing water resources.

Ecological indices or indicators provide a means of assessing the health or status of ecosystems (Jørgensen et al. 2010) by combining several water quality parameters into a single dimensionless value. These indices serve as a tool for environmental management (Xu et al. 2005; Jørgensen et al. 2010; CCME 2017) and as an early warning system for water quality problems (Niemi and McDonald 2004). A variety of ecological indicators exist, each based on different ecosystem parameters, including chemical concentrations (e.g., trophic state index; Carlson 1977), biological parameters such as bioindicator species (Borja et al. 2000) or biodiversity metrics (e.g., Shannon–Wiener index). There are over 20 versions of the Water Quality Index (WQI) used by 35 countries and institutions to evaluate water quality (Uddin et al. 2021). Among them, the Canadian Council of Ministers of Environment Water Quality Index (CCME-WQI) has been applied world-wide to evaluate water quality in rivers, lakes and other sources (Uddin et al. 2021).

In river systems, hydrology can be considered a macro-regulator of ecosystem structure and processes across all scales (Thorpe et al. 2008; Schmutz and Sendzimir 2018). Thus, hydrological patterns determine environmental heterogeneity and, consequently, the concentration of substances along the longitudinal, lateral, and vertical

axes of the river (Thomaz et al. 2007). In turn, discharge also regulates the transport processes of particulate substances (Mayora et al. 2017), with impacts varying across physicochemical variables (Mayora et al. 2013). Depending on the origin of these substances (main channel or floodplain) and the hydrological phase, the relationship between water quality parameters and discharge varies (Preston et al. 1989; Villar and Bonetto 2000; Mayora et al. 2013, 2017). These dynamics highlight the importance of studies examining the relationship between water quality parameters and discharge, as they provide insights into the impact of hydrological regimes on water quality—key aspects for managing and utilizing rivers as multiple-use resources (e.g., drinking water supply, recreation).

Rivers in the agricultural regions of southeastern South America—including the Uruguay River Basin—are exposed to the intensification and conversion of land-uses (Graesser et al. 2015; Modernel et al. 2016; Baeza et al. 2022). The Uruguay River basin has seen an increase in agricultural land area since 1970 (Saurral et al. 2008; Gazzano et al. 2019; Kruk et al. 2023). In particular, the cultivation of rice, wheat and soybeans and, to a lesser extent, livestock has increased (Chalar 2006). The lower Uruguay River can be categorized as eutrophic (average total phosphorus = $70 \mu\text{g l}^{-1}$) (CARU Scientific Committee 2019a, b, c) and presents seasonal algal blooms along the main river channel (Chalar 2009; O'Farrell and Izaguirre 2014; Kruk et al. 2023), often originating in the Salto Grande reservoir and the Negro River reservoirs (González-Piana et al. 2011, 2017; Giannuzzi et al. 2011; O'Farrell et al. 2012; Martínez de la Escalera et al. 2017; Aubriot et al. 2020; Kruk et al. 2021; Alcántara et al. 2022). Some studies also indicate chemical contamination, such as pesticides (Williman et al. 2017; Ernst et al. 2018; Storck et al. 2026), products for pharmaceutical use in humans (Rojo et al. 2019; Storck et al. 2026) as well as metals and organic compounds (Míguez et al. 2010) in biota. Considering that all these studies focus on a single water quality dimension like eutrophication, more integrated comprehensive analysis of water quality and its spatial and temporal variability is needed (García-Alonso et al. 2025; Nogueira et al. 2021).

The main objective of this study was to evaluate the long-term temporal variation of water quality in the mid-lower Uruguay River and their relationship with hydrology. In this context, the following research questions were posed: given the intensification of land use in the Uruguay River basin, is there a deterioration in water quality over time? What is the role of river discharge in the temporal dynamics of water quality and are there site-specific differences in responses? To answer these questions, publicly-available information generated by

the institutions responsible for monitoring and managing the mid-lower Uruguay River was pooled into a single database. Bayesian Generalized Linear Models were applied to evaluate each water quality parameter over time and in relation to discharge, and an integrative water quality index using the CCME-WQI was calculated. This index is a comprehensive approach to evaluate ecosystem health, integrating several parameters and the frequency and magnitude at which they exceed international guidelines for the Uruguay River (CARU 2019). This study seeks to synthesize information into a single, large database of water quality data and evaluate potential drivers of change, ultimately contributing to environmental management strategies for the mid-lower Uruguay River.

Methods

Study area

The Uruguay River is one of the largest rivers in South America and the second largest tributary of the Río de la Plata estuary in terms of discharge (Saurral et al. 2008). The Uruguay River has a length of 1779 km and its basin (between 28°10'S and 37°08'S) covers an area of approximately 356000 km² (Saurral et al. 2008), including the countries of Brazil (53% of the basin surface), Argentina (29%) and Uruguay (18%), and is home to 975000 inhabitants (Achkar et al. 2012; Teixeira De Mello et al. 2025). Based on its geology and hydrography, the Uruguay River can be divided into three sections: upper (from the confluence of Pelotas and Canoas rivers to the mouth of the Peperí River (Yucuma falls) in Brazil, length = 816 km), middle (from Peperí River to Salto Grande reservoir, length = 650 km) and lower (from Salto Grande reservoir to kilometer 0 in Punta Gorda, bordering Uruguay and Argentina, length = 500 km) (Di Persia et al. 1986 and Fig. 1).

Annual rainfall in the Uruguay River Basin averages 1600–1800 mm in the upper basin, 1300–1600 mm in the middle basin and 1000–1300 mm in the lower basin (Krepper et al. 2003). The hydrological regime of the river is driven mainly by precipitation in the upper and middle sections of the basin (Krepper et al. 2003). Discharge varies longitudinally along the main channel in association with slope, declining with decreasing slope from 43 cm km⁻¹ in the upper Uruguay River to 9 and 3 cm km⁻¹ in the middle and lower Uruguay River, respectively. Hydrology of the lower section is mainly regulated by the discharge from the Salto Grande Dam (31°16'S, 57°56'W, average discharge = 5522 m³ s⁻¹, Salto Grande 2026). Currently, the main channel of Uruguay River contains four large hydroelectric dams, and the Uruguay River basin contains in total 187 dams with a water surface area of more than 1 km² (Lehner et al.

2024) with plans to build more than 80 additional dams (Mulligan et al. 2020; Lehner et al. 2024).

Monitoring data

Available water quality data generated by the Uruguay River Management Commission (CARU, Uruguay-Argentina) and hydrological data from the Argentinean National Water Information System (SNIH 2024) were used in this study.

Water quality data were obtained from online reports published by CARU (CARU Scientific Committee 2016, 2017, CARU Scientific Committee 2019a, 2019b, 2019c, 2019d, CARU Scientific Committee 2020 and CARU's *Water Quality and Contamination Control Program* (PROCON) database available online at CARU (2023)). These data included four monitoring programs: *PROCON* with 185 water quality parameters measured from 1987–2005 in 38 sites, *Influence zone of the Orion plant and Guleguaychú River* program with 94 water quality parameters from 2012–2019 in 32 sites, *Initial monitoring plan for the Uruguay River* (PIMI) program with 95 parameters from 2016–2017 in 9 sites, and *Comprehensive monitoring plan for the Uruguay River for the shared section* (MIRU) with 181 parameters from 2018–2019 in 28 sites.

From the network of more than 48 sites in the CARU monitoring programs (including nearshore sites and in the central channel), three sampling sites—labelled by their distance upstream from the mouth (Km 0)—were selected in the mid-lower section of the Uruguay River: Monte Caseros–Bella Unión (RUY480, 30°13'S, 57°37'W), Fray Bentos–Guleguaychú (RUY90, 33°07'S, 58°22'W) and La Concordia (RUY40, 33°34'S, 58°28'W) (Fig. 1). The sites selected were located in the central channel of the river to represent overall fluvial dynamics rather than potential localized inputs from the river margins (Miguez 2007). These three sites were selected because those were the only sites located on the main channel consistently sampled along all the monitoring periods.

A series of criteria were developed to select variables for analysis, including (1) measurements available in all monitoring programs, and (2) parameters with less than 40% missing data (NA) across the entire study period. As the detection limits varied over time depending on the monitoring programs (Table S1), values reported as below the detection limit (DL) were replaced with the minimum DL reported. To remove potential errors, for all variables, values < Percentile 0.5 or > Percentile 99.5 for each site, were considered as outliers and removed from the dataset. Sampling frequency ranged from semi-annual to monthly, varying among sites and monitoring programs. When more than one sampling was performed

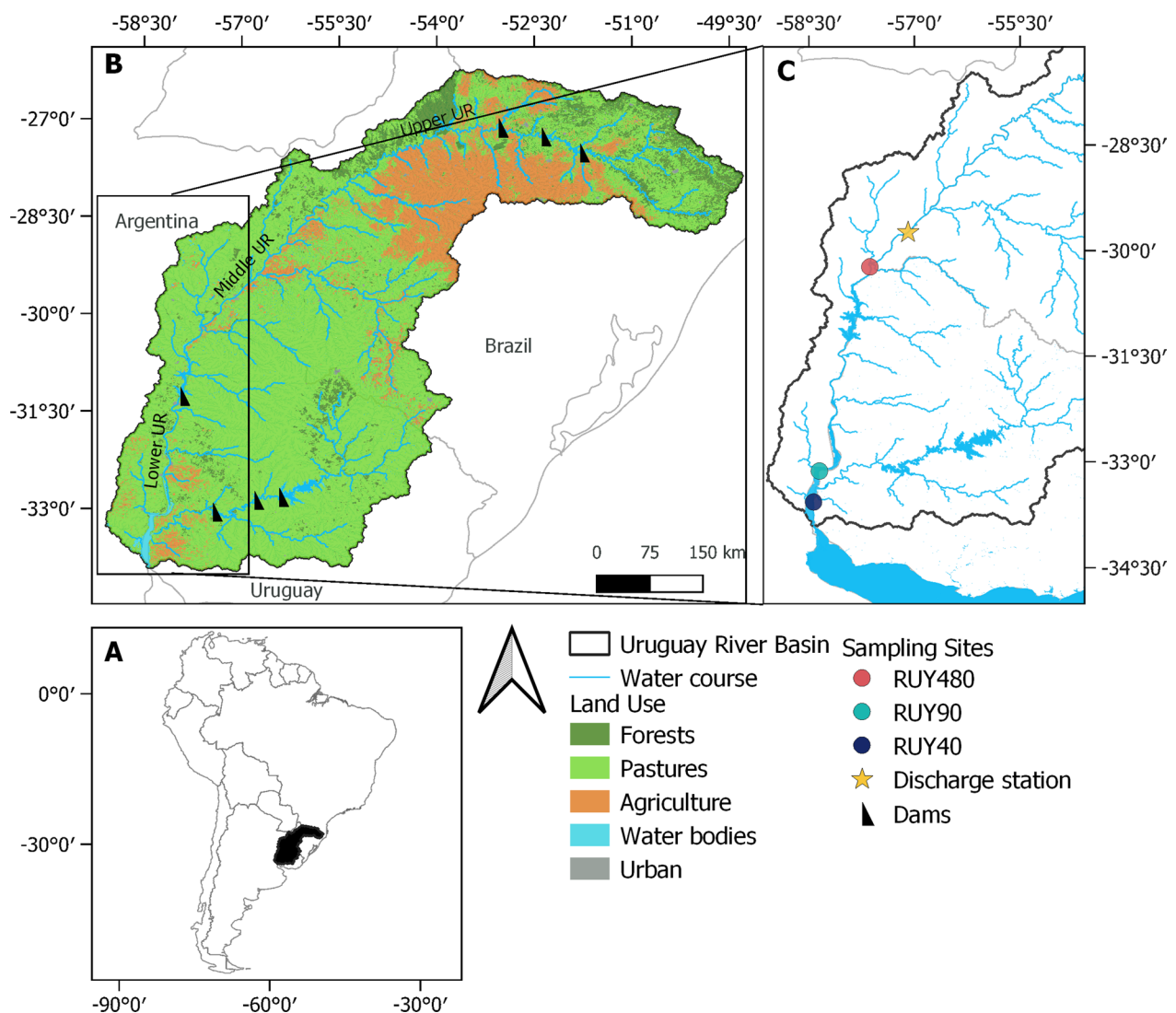


Fig. 1 **A** Map of the Uruguay River Basin in South America. **B** Land Use/Land Cover (from MODIS 2018) in the Uruguay River basin. Triangles show the hydroelectric dams in the main channel of the Uruguay River: Machadinho, Itá, Foz do Chapecó and Salto Grande and in the Negro River. **C** Map of the study area (mid-lower Uruguay River) showing the locations of water quality sampling sites (circles: RUY480: Monte Caseros–Bella Unión, RUY90: Fray Bentos–Guaquaychú and RUY40: La Concordia) and Paso de Los Libres hydrometric station (yellow star)

on the same day (e.g., morning and afternoon), the data were averaged. When sampling was performed on consecutive days, a single date was selected to avoid bias in the database, using the following criteria: (a) selecting the date with the highest number of measured water quality parameters, or (b) selecting the middle date. Variables were grouped into six types: physicochemical, nutrients, metals, microbiological, pesticides, and persistent, bioaccumulative and toxic chemicals (PBTs).

Daily discharge (Q , $\text{m}^3 \text{s}^{-1}$) data from 1985–2020 of the Uruguay River were obtained from publicly available platforms of the SNIH (data available online at SNIH website (SNIH 2024)). The Paso de los Libres (Argentina)

hydrometric station, located 570 km upstream from Km 0 was selected for models as it was upstream of all sites (Fig. 1). It is important to highlight that there was a strong linear correlation between Q at Paso de los Libres and Q at other hydrometric station downstream of the Salto Grande Dam (Concordia, located 320 km upstream from Km 0) (Pearson correlation = 0.91, $p < 0.05$).

Discharge percentile was determined and each water quality sampling was classified according to three hydrological conditions: below-average discharge ($< Q_{25}$), average discharge (Q_{25-75}) and above-average discharge ($> Q_{75}$) (Fig. S1). For each hydrological condition, Bayesian models were developed (see below) to understand

the effect of year and discharge on the nutrients concentration.

Data analysis

Descriptive analysis and comparison with regulations

To assess differences in the water quality parameters among sites for the entire study period, particularly to evaluate potential upstream or downstream trends, non-parametric Kruskal–Wallis tests (K-W, $p < 0.05$) followed by Dunn post hoc tests ($p < 0.05$) were used. To evaluate differences in water quality parameter values over time, concentrations across values grouped into decades, where 1980s = 1987–1989; 1990s = 1990–1999, 2000s = 2000–2009 and 2010s = 2010–2019 were compared using the same approach. The grouping of data into decades was implemented in order to have sufficient n for the time comparison in this analysis.

The water quality parameters values were compared with threshold guideline values (Table 1, CARU 2019) to evaluate whether they remained within recommended limits over time. To analyze the frequency in which a parameter did not comply with guideline value per year, the “frequency of regulatory non-compliance” was calculated for each parameter as the number of times that the parameter fails to comply with the guideline values divided by the number of times it was measured in the year. Standardizing these values by the number of times the parameter measured in the year allows comparison among years. Since the frequency of values of non-compliance for all parameters did not follow a normal distribution (Shapiro–Wilk test, $p < 0.05$ for all parameters), a Spearman correlation test was used to estimate the correlation between the frequency of regulatory non-compliance and the year.

Bayesian modeling approach

To assess the effects of time and discharge on water quality parameters, a Bayesian modelling approach (Bayesian generalized linear models, GLMs) was applied. For Bayesian GLMs, only data $> DL$ were considered. For each water quality parameter, different distributions were evaluated using maximum-likelihood fitting for the response variable and considered the one that best fits each parameter (Table S4) (fitdistrplus package, Delignette-Muller and Dutang 2015). All models were built using the following structure:

$$\text{Water quality parameter} = \text{year} \times \text{site} + \log(Q) \times \text{site}$$

where year, $\log(Q)$ and site were included as fixed effects representing population-level predictors. Year was treated as a numeric variable ranging from 1 to 33, with 1 corresponding to 1987 and 33 to 2019, to improve model convergence. $\log(Q)$ was centered and standardized

(Schielzeth 2010). This fitted model allows us to estimate the effect of year and $\log(Q)$ per site (Table S4). The model was fitted using Stan (Stan Development Team 2023) through its R interface brms package (Bürkner 2017, 2018, 2021). For each model, a prior sensitivity analysis between weak and strong prior distributions following Inchausti (2022) was performed and used strong priors (Table S4). Three chains for 6000 iterations were run, leaving 1000 warm-up. The minimum effective sample size (i.e. number of statistically independent estimates of each parameter) was 1016 and maximum Rhat was 1.00, indicating convergence of the 3 chains (Vehtari et al. 2017; Inchausti 2022). For each model, Bayesian R^2 was calculated to determine the proportion of variation in the water quality parameter explained by the model (Gelman et al. 2019; Inchausti 2022). Based on posterior predictive checks and residual analyses using randomized quantile residuals (relation between the mean of Pearson residuals and fitted values of each explanatory variable, quantile–quantile plot of the randomized quantile residuals, the Pareto k cross-validation statistics for each data point and the quantile–quantile plot of mean group effects of each site) all models showed a very good to excellent goodness of fit to the data. The models were applied to all variables except pesticides and mercury due to a limited number of values $> DL$.

As pesticides frequently array results below detection limits and could not be modeled with Generalized Linear Models (GLMs) due to data distribution (i.e. excess of values $< DL$), Spearman correlations ($p < 0.05$) were made between the concentration of each pesticide and the year to evaluate trends over time.

As nutrients are of particular interest in water quality in the Uruguay River, and their concentrations have frequently been shown to depend on river discharge scenarios (Mayora et al. 2017), a Bayesian GLM for each hydrological condition separately was also developed. The model structure was:

$$\text{Water quality parameter} = \text{year} + \log(Q)$$

where the slopes for year and discharge were estimated with all sites together. The same specifications mentioned above were used to fit the models for each hydrological condition.

Water quality index

A Water Quality Index (WQI) is based on an aggregation function that allows analysis of large temporal and spatial water quality datasets to produce a single value (WQI) that indicates ecosystem status. These indices are based on local water quality criteria, for example, current regulations or opinions of a panel of experts on the subject (Uddin et al. 2021). As many indices refer to local

Table 1 Water quality parameters evaluated in this study by variable type from 1987–2018

Water quality parameter	Units	Min	Median	Max	N	N > DL	Lower limit	Upper limit
<i>Physicochemical</i>								
Alkalinity	mg l ⁻¹	10	22	52	146	146	18	46
Calcium	mg l ⁻¹	3.5	7.3	39.0	212	212		9
Chlorides	mg l ⁻¹	< DL	2.0	14.3	204	192		2
Conductivity	μS cm ⁻¹	15.0	60.0	170.0	231	231	40	100
Cyanide*	mg l ⁻¹	< DL	0.003	0.003	40	0		5
Fluorides*	mg l ⁻¹	< DL	0.15	0.15	40	2		0.12
Hardness	mg l ⁻¹	8.0	24.5	67.5	228	228	15	40
Dissolved oxygen	mg l ⁻¹	6.2	8.3	10.8	214	214	5.0	15.0
Magnesium	mg l ⁻¹	1.10	2.22	13.9	163	163		2.5
Oils and grease*	mg l ⁻¹	< DL	4.5	9.5	40	19		1
pH		6.3	7.3	8.0	231	231	6.5	9.0
SAAM*	μg l ⁻¹	3.7×10 ⁻⁵	7.27×10 ⁻⁵	0.0002	37	37		280
Sulfates	mg l ⁻¹	< DL	1.9	25	207	154		2.5
Sodium	mg l ⁻¹	< DL	2.8	15.0	196	161		4
Temperature	°C	12	20.7	30	232	232	13	30
Turbidity	NTU	10.8	29.1	146.7	209	209		65
Biological oxygen demand (BOD ₅)	mg l ⁻¹	< DL	1.3	16.0	200	110		3
Chemical oxygen demand (COD ₅)	mg l ⁻¹	< DL	15.0	69.0	228	126		12
Fixed suspended solids (FSS)	mg l ⁻¹	< DL	10.0	80	213	143		20
Total dissolved solids (TDS)	mg l ⁻¹	< DL	83.0	214.0	158	153		
Total suspended solids (TSS)	mg l ⁻¹	< DL	11.0	90	214	177		30
<i>Nutrients</i>								
Nitrate	mg l ⁻¹	0.07	0.63	1.87	212	212		10
Nitrite	mg l ⁻¹	< DL	0.0013	0.05	216	122		0.06
Kjeldahl nitrogen	mg l ⁻¹	< DL	0.41	1.82	208	202		0.5
Total nitrogen (TN)	mg l ⁻¹	0.27	1.06	2.19	202	202		
Total phosphorus (TP)	mg l ⁻¹	< DL	0.06	0.38	219	213		0.1
Orthophosphates*	mg l ⁻¹	< DL	0.041	0.083	32	32		0.04
<i>Metals</i>								
Arsenic	mg l ⁻¹	< DL	0.0006	0.0015	149	75		0.0015
Boron*	mg l ⁻¹	0.1	0.1	0.1	37	37		0.37
Cadmium	mg l ⁻¹	< DL	0.00009	0.002	225	77		0.0002
Copper	mg l ⁻¹	< DL	0.005	0.032	216	168		0.01
Chromium	mg l ⁻¹	< DL	0.002	0.072	233	176		0.0089
Chromium(VI)*	mg l ⁻¹	< DL	0.007	0.016	40	13		0.001
Iron	mg l ⁻¹	0.14	1.7	5.51	173	173		1.8
Lead	mg l ⁻¹	< DL	0.00085	0.009	231	166		0.007
Mercury	mg l ⁻¹	< DL	0.00001	0.00002	222	7		0.00005
Nickel*	mg l ⁻¹	< DL	0.009	0.028	40	7		0.025
Tin*	mg l ⁻¹	0.001	0.001	0.001	37	37		0.02
Selenium*	mg l ⁻¹	0.005	0.005	0.005	37	37		0.005
Zinc	mg l ⁻¹	< DL	0.01	0.1	192	159		0.037
<i>Microbiological</i>								
Thermotolerant coliforms	UFC 100 ml ⁻¹	< DL	77.2	1420	178	174		
Cyanobacterial abundance*	cel ml ⁻¹	< DL	12	537	40	38		5000
Chlorophyll-a*	μg l ⁻¹	0.1	3	11.8	40	40		10
<i>Pesticides</i>								
Alachlor*	μg l ⁻¹	< DL	0.005	0.005	38	38		0.3

Table 1 (continued)

Water quality parameter	Units	Min	Median	Max	N	N > DL	Lower limit	Upper limit
Aldrin	µg l ⁻¹	< DL	0.0017	0.0043	212	11		0.005
Atrazine*	µg l ⁻¹	< DL	0.016	0.205	30	30		1.8
Carbofuran*	µg l ⁻¹	< DL	0.004	0.084	40	4		1.2
Chlorpyrifos*	µg l ⁻¹	< DL	0.033	0.033	40	5		0.03
DDT	µg l ⁻¹	< DL	0.003	0.009	183	10		0.001
Dicamba*	µg l ⁻¹	< DL	0.024	0.033	40	2		10
Dichlorvos*	µg l ⁻¹	< DL	0.014	0.014	40	1		0.0006
Dieldrin	µg l ⁻¹	< DL	1.7	3.0	205	7		0.005
Endosulfan	µg l ⁻¹	< DL	0.0006	0.006	147	27		0.02
Endrin	µg l ⁻¹	< DL	0.0003	0.0003	212	0		0.004
Glyphosate + AMPA*	µg l ⁻¹	< DL	0.015	0.032	40	12		800
Heptachlor	µg l ⁻¹	< DL	0.0012	0.002	216	11		0.01
Heptachlor epoxide	µg l ⁻¹	< DL	0.001	0.004	197	7		0.01
Lindane	µg l ⁻¹	< DL	0.0012	0.007	211	28		0.01
Parathion	µg l ⁻¹	< DL	0.001	0.001	140	0		0.004
2,4,5-TP*	µg l ⁻¹	< DL	0.017	0.017	40	1		2
2,4-D*	µg l ⁻¹	< DL	0.028	0.052	40	2		4
<i>PBTs</i>								
Total hydrocarbons	µg l ⁻¹	< DL	12.7	6800	165	43		
Phenols	µg l ⁻¹	< DL	2.2	25.5	159	62		1

Units of measurement, median, minimum (min) and maximum (max) values, amount of data different from NA (N), amount of data greater than the detection limit (N > DL) and the lower and upper limits guideline values for the Uruguay River (CARU 2019) are shown. PBTs: Persistent, bioaccumulative and toxic chemicals. Asterisks (*) represents the variables that were only analyzed for the recent period (2018–2019, MIRU program). The specific values < DL were in Table S1

water quality standards (Xu et al. 2001) and are often generated with different parameters, calculations and aggregation formulas (Lumb et al. 2011), water quality categories allow for comparison among different ecosystems (CCME 2001; Almeida et al. 2012).

In this study, the CCME-WQI was used to estimate the water quality of the mid-lower Uruguay River. The CCME-WQI has several advantages over other indexes: it can select the number of parameters to be included, compare with the regulatory values of each location, monitor the water quality status of the site over time or compare between locations, and has been adopted by several countries and institutions (Uddin et al. 2021; Fortes et al. 2023).

The CCME-WQI is based on a combination of three factors: Scope (the percent of variables where thresholds are exceeded at least once); Frequency (the percent of individual tests in which thresholds are exceeded); and Amplitude (the amount by which the thresholds are exceeded). These three factors are combined to produce a single CCME-WQI value (between 0 and 100) which is then converted to a ranking of quality (0–44 = Poor; 45–64 = Marginal; 65–79 = Fair; 80–94 = Good; 95–100 = Excellent) intended to describe overall water quality. In this study, the regulations on the lower Uruguay River (CARU 2019) were used as reference to calculate

the CCME-WQI. The 30-year CCME-WQI (1987–2019 period) was calculated for two consecutive years to ensure data representation. The 30-year CCME-WQI considered water quality parameters with reference values from all Variable Types (Table 1, CARU 2019), but excluding variables with high correlation (Spearman rank correlation coefficient < |0.7|) to avoid redundancy (Dormann et al. 2013). Based on the CCME-WQI guide, it is recommended to use between 8 and 20 water quality parameters to calculate the CCME-WQI (CCME 2001). The water quality parameters included in the 30-year CCME-WQI were conductivity, pH, and the concentration of: sulfates, lindane, heptachlor, lead, hardness, COD₅, chlorides, cadmium, TSS, heptachlor epoxide, dissolved oxygen, nitrite, chromium, nitrate, total phosphorus, and endrin. The importance of each water quality parameter in the CCME-WQI results was determined by a Random Forest with 500 trees (Liaw and Wiener 2002) using IncNodePurity as a measure of the total decrease in a node impurity that results from splits over the variable and averaged over all trees (James et al. 2013).

Based on more recent monitoring data from the MIRU program (June 2018–December 2019, CARU Scientific Committee 2019a, b) with 181 water quality parameters, a CCME-WQI was calculated for each Variable Type per semester (second semester of 2018 and both semesters of

2019). For each Variable Type, water quality parameters were selected that met the following conditions: have less than 50% of missing data (NA), have at least one value > DL, have reference values in Uruguay River regulations (CARU 2019), and not being correlated with other parameters (Spearman rank correlation coefficient < |0.7|, Dormann et al. 2013). To calculate CCME-WQI by Variable Type, 15 physicochemical variables (oils and grease, calcium, cyanide, chlorides, conductivity, BOD₅, COD₅, fluorides, magnesium, dissolved oxygen, pH, SAAM, sodium, sulfates, and turbidity), 13 metals (arsenic, boron, cadmium, copper, chromium, chromium(VI), tin, iron, mercury, nickel, lead, selenium and zinc), and 9 pesticides (alachlor, atrazine, carbofuran, chlorpyrifos, dicamba, dichlorvos, 2,4,5-TP, 2,4-D, and glyphosate + AMPA) were selected. In addition, a CCME-WQI that accounted for trophic state was calculated, which included 5 nutrients (total phosphorus, orthophosphates, nitrite, nitrate, and Kjeldahl nitrogen) and 2 microbiological parameters related to phytoplankton (cyanobacterial abundance and chlorophyll-*a*). A Random Forest was used to determine the importance of each variable in explaining each calculated CCME-WQI. Based on the water quality parameters with the highest importance in each CCME-WQI, and fecal coliforms, phenols and AOX (not able to be grouped in any of the defined Variable Types previously), a general CCME-WQI was calculated for the recent period (2018–2019) including all Variable Types (Physicochemical variables: sulphates, magnesium, turbidity, BOD₅, dissolved oxygen and oils and grease, Nutrients: total phosphorus, orthophosphates, nitrate and nitrite, Microbiological: cyanobacterial abundance and fecal coliforms, Metals: zinc, chromium(VI) and iron; Pesticides: atrazine, carbofuran and chlorpyrifos, and, PBTs: phenols and AOX). A Random Forest was also used to determine the impact of different variables on overall CCME-WQI.

The CCME-WQI was calculated using the *wqindex* package (Thorley et al. 2018). This R package calculates and plots CCME-WQI with the specified guidelines values.

All analyses in this study were performed in the R environment (R Core Team 2023).

Results

Characterization of water quality parameters and comparison with regulations

The final 30-year water quality dataset included 246 observations of 42 variables from 1987 to 2019 (Table 1), with gaps in monitoring from 1996 to 1998 and 2005–2012. The number of observations per sampling site over the entire period was uneven (RUY480 = 72, RUY90 = 131 and RUY40 = 43). The MIRU program database for the recent period (2018–2019) included 21 new water quality parameters and 40 observations (Table 1).

Among sites, significant differences (K-W test, $p < 0.05$) were found in the physicochemical parameters alkalinity, conductivity, hardness, COD₅, dissolved oxygen, pH, sodium, FSS, TSS, and turbidity (Fig. S2 and Table S2). In particular, alkalinity, conductivity, hardness, dissolved oxygen, pH, and sodium increased downstream (RUY480 < RUY90 < RUY40), while physicochemical parameters related to solids (FSS, TSS, and turbidity) decreased between mid to lower Uruguay River (RUY480 > RUY90 and RUY40) with RUY480 having significantly higher values than all other sites, but without significant differences between RUY90 and RUY40 (Fig. S2). Differences among sites were found for the nutrients Kjeldahl nitrogen, nitrite and total nitrogen; the metals arsenic, chromium and lead; the pesticides aldrin, dieldrin, endosulfan, heptachlor, heptachlor epoxide and lindane (K-W test, $p < 0.05$, Table S2). Phenols and thermotolerant coliforms also showed significant differences among sites (K-W test, $p < 0.05$, Table S2). Particularly, Kjeldahl nitrogen, total nitrogen, chromium, lead, aldrin, dieldrin, heptachlor, heptachlor epoxide, and lindane decreased downstream (Fig. S2). RUY90 showed lower average values of nitrites, COD₅ and phenols, and the higher average values of arsenic and endosulfan than the other sites (Dunn test, $p < 0.05$, Fig. S2).

Water quality parameters across decades

The pesticides aldrin, DDT, dieldrin, heptachlor, heptachlor epoxide and lindane; and the nutrients Kjeldahl nitrogen and total nitrogen showed significant differences across decades (K-W test, $p < 0.05$; Fig. 2 and Table S3). All these pollutants showed higher average values in the

(See figure on next page.)

Fig. 2 Water quality parameters across decades (box-colors represent Variable Types: physicochemical, nutrients, microbiological, metals, pesticides and PBTs). The box represents the 25th and 75th percentiles, the black line is the median, whiskers are the 1.5 * inter-quantile range (1.5IQR), black points are values outside the 1.5IQR, grey dots represent data points and red dashed lines are the threshold guideline values (CARU 2019). Asterisks represent significant differences among decades (*: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$; ****: $p < 0.0001$). There is no magnesium data for 1980s. Endrin and parathion are not shown, as their values were consistently below the detection limit, and arsenic was also excluded because its maximum concentration (RUY90 2012–09–17, concentration = $0.00154 \mu\text{g l}^{-1}$) was 1000 times lower the threshold guideline value ($15 \mu\text{g l}^{-1}$, CARU 2019). Y-axis scales are independent

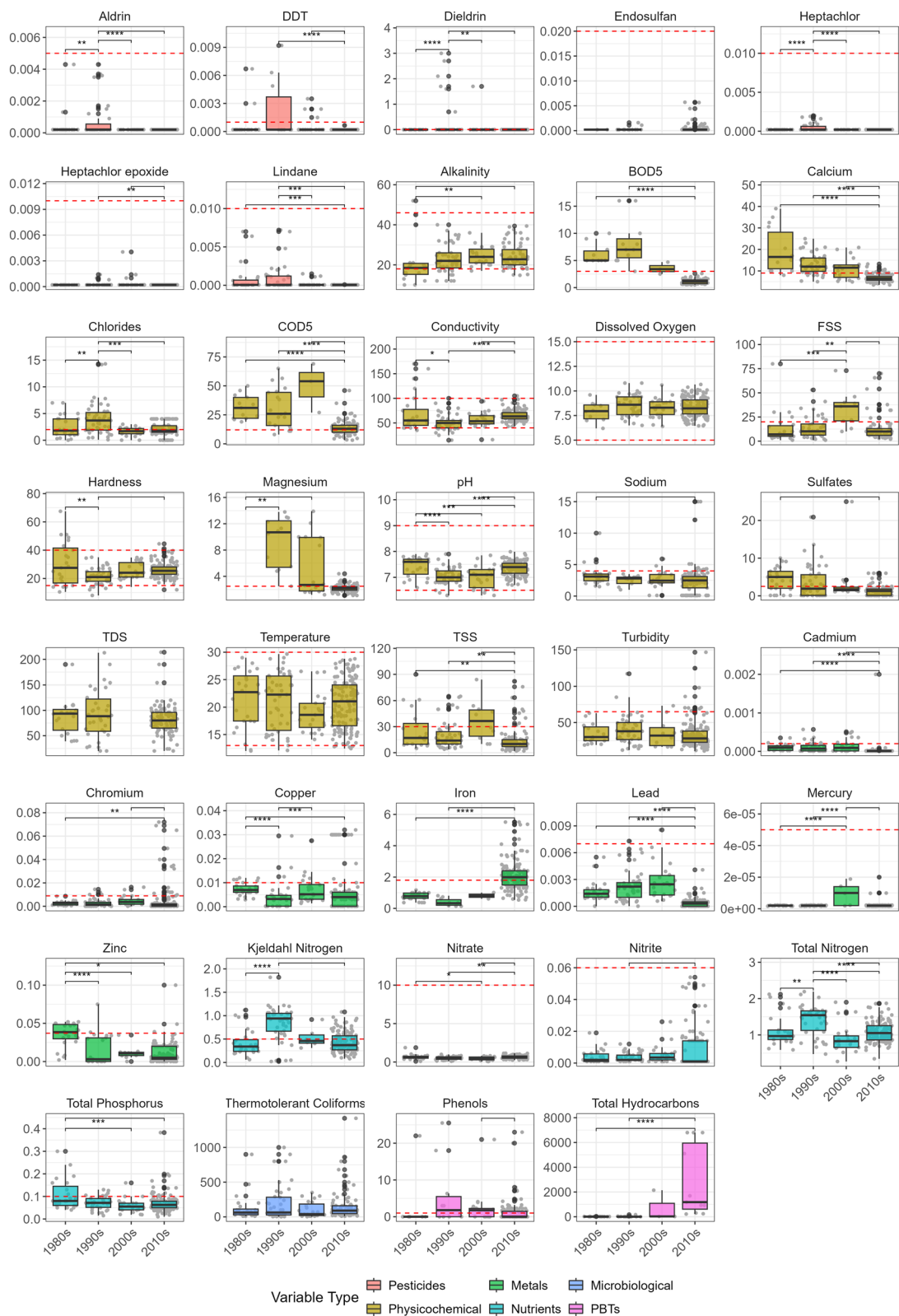


Fig. 2 (See legend on previous page.)

1990s than in the other decades (Dunn test, $p < 0.05$; Fig. 2). Furthermore, nitrate concentrations were significantly lower in the 2000s than in the other decades (Dunn test, $p < 0.05$; Fig. 2). The average total phosphorus concentration was higher in the 1980s compared to the 2000s and 2010s (Dunn test, $p < 0.05$). Although a trend towards decreasing total phosphorus concentration was observed over time (Fig. 2), the maximum values were recorded in the most recent decade (0.38 mg l^{-1} in 2019 in RUY480).

The physicochemical variables hardness and pH also showed significant differences across decades (K-W test, $p < 0.05$; Table S3) with lower average values in the 1990s than in 1980s and 2010s (Fig. 2). Magnesium and calcium showed a decreasing trend over time (Fig. 2). Regarding the concentration of metals in water, iron showed higher average in the 2010s than in the 1980s and 1990s, while lead and cadmium had the lowest average over the same period (Dunn test, $p < 0.05$; Fig. 2). Regarding the parameters grouped within the PBTs Variable Type, the average of total hydrocarbons increased over time with the highest concentrations in the 2010s. Finally, endosulfan, dissolved oxygen, TDS, temperature, turbidity, and thermotolerant coliforms did not show significant differences across decades ($p > 0.05$; Fig. 2 and Table S3).

Regulatory compliance over time

Seventy percent of the parameters that have a threshold guideline value in the current regulation (CARU 2019) failed to comply with the guideline value at least once in the 30 years studied (Fig. 2 and Fig. 3). Chlorides and iron were the parameters that most frequently exceeded the threshold guideline values, with 45% of the measurements surpassing threshold guideline values (Fig. 2 and Fig. 3). The physicochemical variables calcium (maximum concentration = 39 mg l^{-1} in RUY90 1988–06–06), chlorides (maximum concentration = 14.3 mg l^{-1} in RUY90 1991–12–09), magnesium (maximum concentration = 13.9 mg l^{-1} in RUY40 2004–07–06), sulfates (maximum concentration = 25 mg l^{-1} in RUY90 2001–08–14), sodium (maximum concentration = 15 mg/L in RUY40 2018–09–04), turbidity (maximum = 146.7 NTU in RUY480 2017–05–03) and BOD_5 (maximum concentration = 16.0 mg l^{-1} in RUY90 1991–09–30), the metals cadmium (maximum concentration = 0.002 mg l^{-1} in RUY90 2017–11–15) and zinc (maximum concentration = 0.1 mg l^{-1} in RUY480 2017–08–08), dieldrin (maximum concentration = $3.0 \text{ } \mu\text{g l}^{-1}$ in RUY480 1993–12–03), phenols (maximum concentration = $25.5 \text{ } \mu\text{g l}^{-1}$ in RUY480 1992–12–13) and Kjeldahl nitrogen (maximum concentration = 1.82 mg l^{-1} in RUY480 1992–04–03) showed values that exceeded the threshold guideline values by more than an order of magnitude (Table 1).

Aldrin, endosulfan, endrin, heptachlor, heptachlor epoxide, lindane, parathion, arsenic, mercury, nitrate, nitrite, and dissolved oxygen remained consistently within regulatory compliance values over the study period (Table 1 and Fig. 2). Iron concentrations exceeded the threshold guideline value (1.8 mg l^{-1} , CARU 2019) only from 2012 onwards, and thermotolerant coliforms surpassed the threshold guideline value for total coliforms (1000 CFU , CARU 2019) only once, in 2018.

The frequency of regulatory non-compliance for DDT (Spearman's correlation: $\rho = -0.63$), BOD_5 ($\rho = -0.63$), calcium ($\rho = -0.55$), conductivity ($\rho = -0.51$), magnesium ($\rho = -0.54$) and sulfates ($\rho = -0.49$) had a significant negative correlation with year ($p < 0.05$ for all variables); while the frequency of regulatory non-compliance for iron had a positive correlation with year ($\rho = 0.88$, $p = 3.6 \times 10^{-5}$) (Fig. 3). The frequency of regulatory non-compliance for dieldrin, alkalinity, chlorides, COD_5 , FSS, hardness, pH, sodium, TSS, turbidity, cadmium, chromium, copper, lead, zinc, Kjeldahl nitrogen, total phosphorus, and phenols had no correlation with year (Fig. 3).

Water quality parameters modeled across time and hydrological variability

In general, the response of water quality parameters to year and discharge varied across sites, with most effects observed only at site RUY480 (Table S4). At RUY480, conductivity, pH, copper, chromium, and total hydrocarbons increased over time, while sulfates, turbidity, FSS, TSS, COD_5 , Kjeldahl nitrogen, and phenols decreased over time (Fig. 4 and Table S4). At all three sites, calcium, BOD_5 , and magnesium decreased over time, while iron and nitrites increased (Fig. 4 and Table S4). Zinc decreased over time only at RUY90 (Fig. 4 and Table S4).

At RUY480, chlorides, conductivity, pH, sodium, magnesium, and total hydrocarbons decreased with discharge, while FSS, TDS, TSS, nitrate, total nitrogen, and total phosphorus increased with discharge (Fig. 5 and Table S4). Turbidity increased with discharge at RUY480 and RUY90 (Fig. 5 and Table S4). Neither time nor discharge had any effect on alkalinity, hardness, dissolved oxygen, arsenic, cadmium, lead, and thermotolerant coliforms (Table S4).

Negative correlations over time (years) were found for aldrin (Spearman's correlation: $\rho = -0.25$, $p < 0.05$), dieldrin ($\rho = -0.2$, $p < 0.05$), DDT ($\rho = -0.25$, $p < 0.05$), endosulfan ($\rho = -0.46$, $p < 0.05$), heptachlor ($\rho = -0.1$, $p < 0.05$), heptachlor epoxide ($\rho = -0.14$, $p < 0.05$) and lindane ($\rho = -0.4$, $p < 0.05$) for the 30-year period (Fig. S3). In the case of heptachlor and endosulfan, a significant positive correlation with year was detected for the period 1987–2005 until its use was banned (Spearman's correlation: $\rho = 0.2$, $p < 0.05$ and $\rho = 0.56$, $p < 0.05$, for

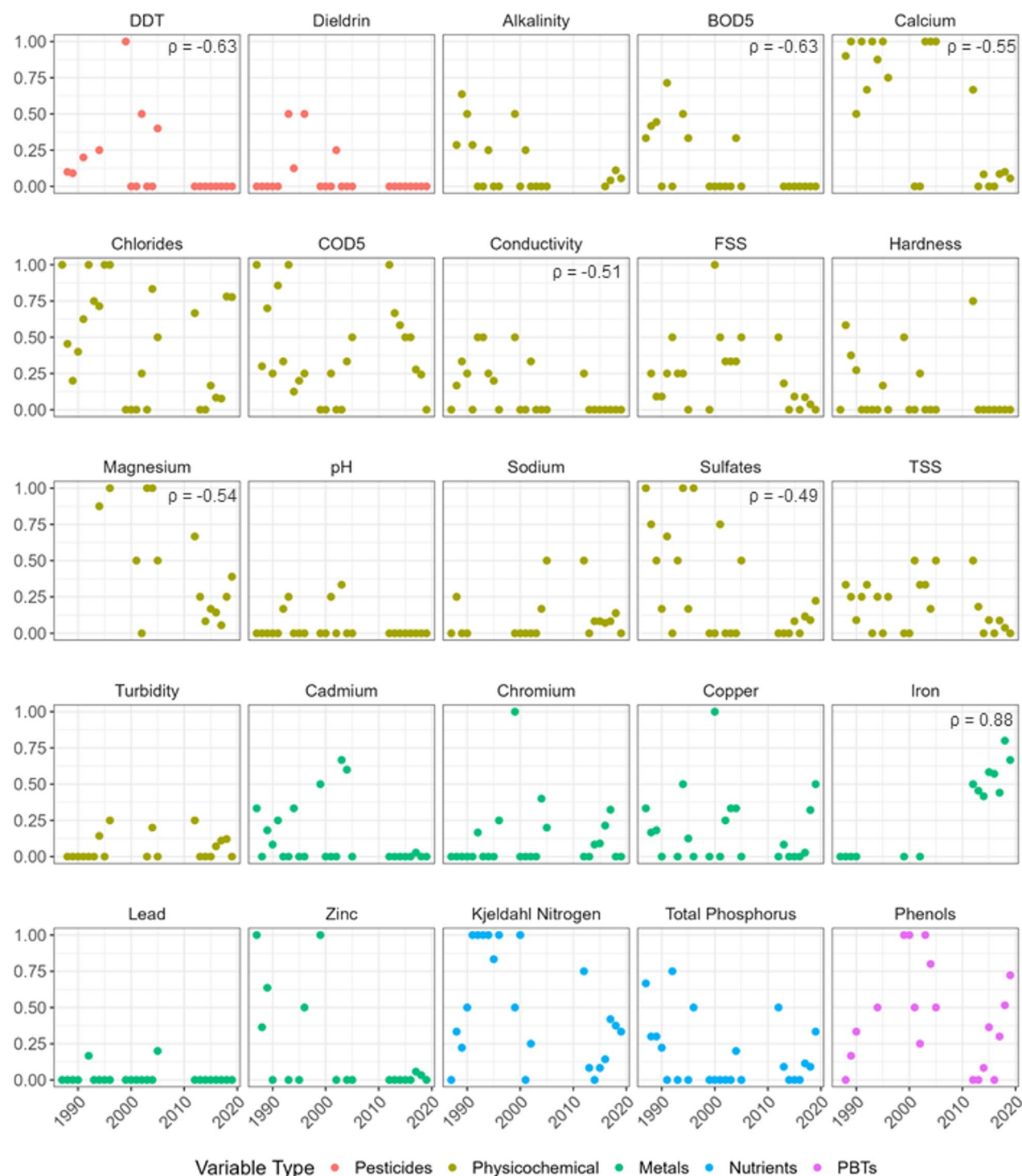


Fig. 3 Frequency of regulatory non-compliance values for each water quality parameter per year. Dot colors represent Variable Types: physicochemical, nutrients, metals, pesticides and PBTs. The Rho value (ρ) represents the significant results of the Spearman correlation test ($p < 0.05$) between the frequency of regulatory non-compliance values for each water quality parameter and year. Only variables with at least one value over or under threshold guideline value are shown

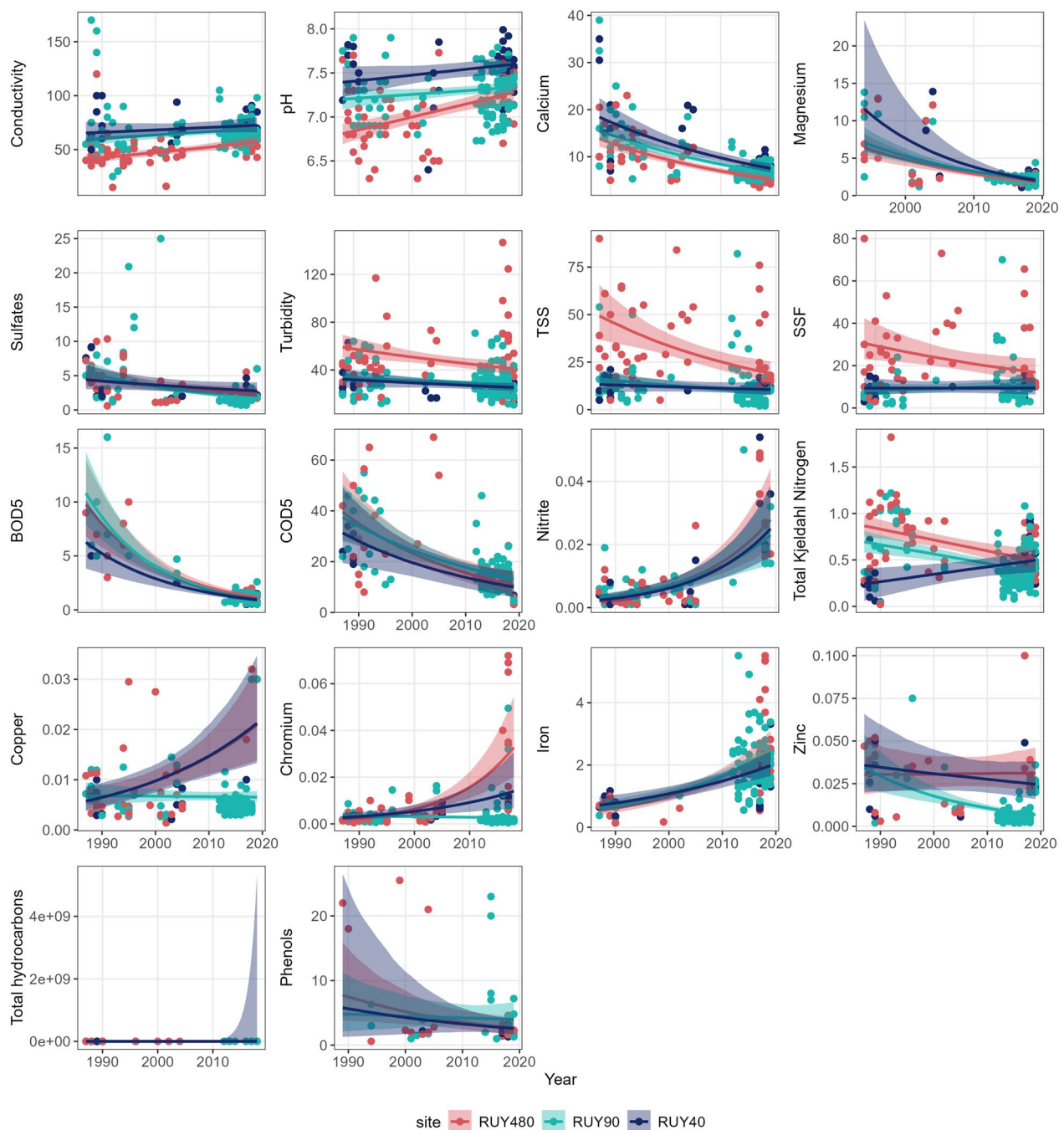


Fig. 4 Time conditional predictive curves of Bayesian GLM ($\text{water quality parameter} = \text{site} \times \text{year} + \text{site} \times \log(Q)$) for each water quality parameter. Colors indicate sites

heptachlor and endosulfan, respectively), while significant negative correlations were observed when considering the 30-year period.

Nutrient concentrations showed different trends across discharge and year depending on hydrological condition. For Q_{25} , total nutrients (TP and TN), nitrates and nitrites increased with discharge (pend > 0.05; $R^2 > 0.2$),

while total nutrient concentrations were not related to discharge in Q_{25-75} and Q_{75} (Fig. 6 and Table S5). Nitrate and Kjeldahl nitrogen concentration increase with discharge in Q_{75} (Fig. 6 and Table S5). Nitrite concentration increased with year in the three hydrological conditions (Fig. 6 and Table S5) with a good fit ($R^2 > 0.5$). At the same time, Kjeldahl nitrogen, total nitrogen and total

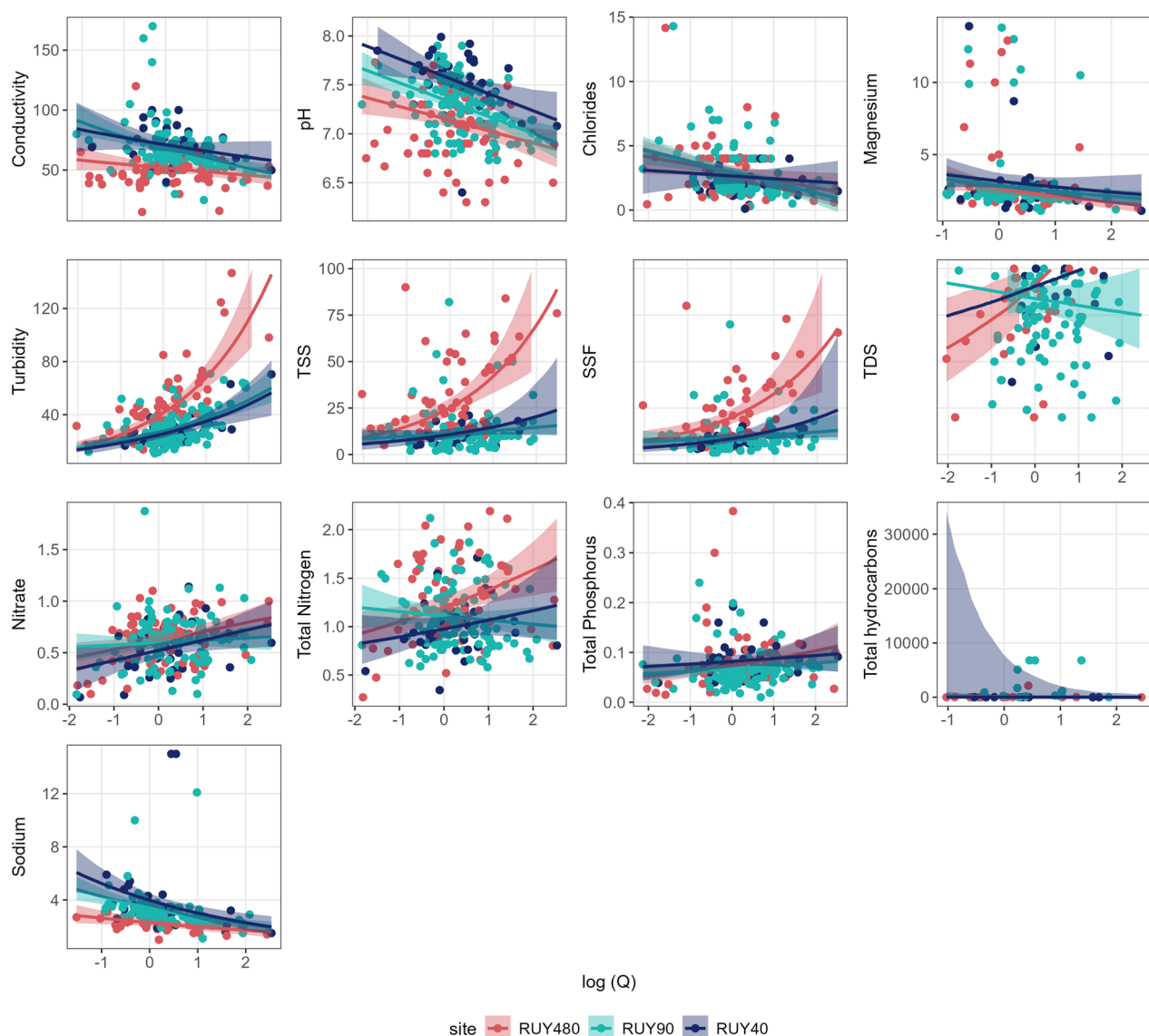


Fig. 5 Discharge conditional predictive curves of Bayesian GLM ($\text{water quality parameter} = \text{site} \times \text{year} + \text{site} \times \log(Q)$) for each water quality parameter. Colors indicate sites. The turbidity graph's y-axis was truncated at 150 for visualization purposes, but the RUY480's credible interval for turbidity versus discharge has its maximum value at 520

phosphorus concentrations decreased with time in Q_{25-75} (Table S5).

Water quality index

CCME-WQI over 30 years

The 30-year CCME-WQI index showed values ranging from Poor to Good among all sites. The 30-year CCME-WQI had the lowest values ($\text{WQI} = 42.4$, Category = Poor) in 1993–1994 and showed an improvement in water quality from 2000 onwards, fluctuating between Fair and Good (Fig. 7A). The Random Forest model had a mean residual error of 91.7 and explained 58.5% of the

variance. The most important variables in the 30-year CCME-WQI calculation were dieldrin, conductivity, sulfates, and lindane (Fig. 7B).

CCME-WQI in recent period

The CCME-WQI of the physicochemical parameters for the recent period showed values between Fair and Good (between 69.0 and 82.9; Fig. 8A). All the physicochemical variables included in the CCME-WQI calculation, except cyanide (all the values measured were $< \text{DL}$, Table 1), contributed to the significance of the CCME-WQI calculation (Fig. S4A). The CCME-WQI of the trophic

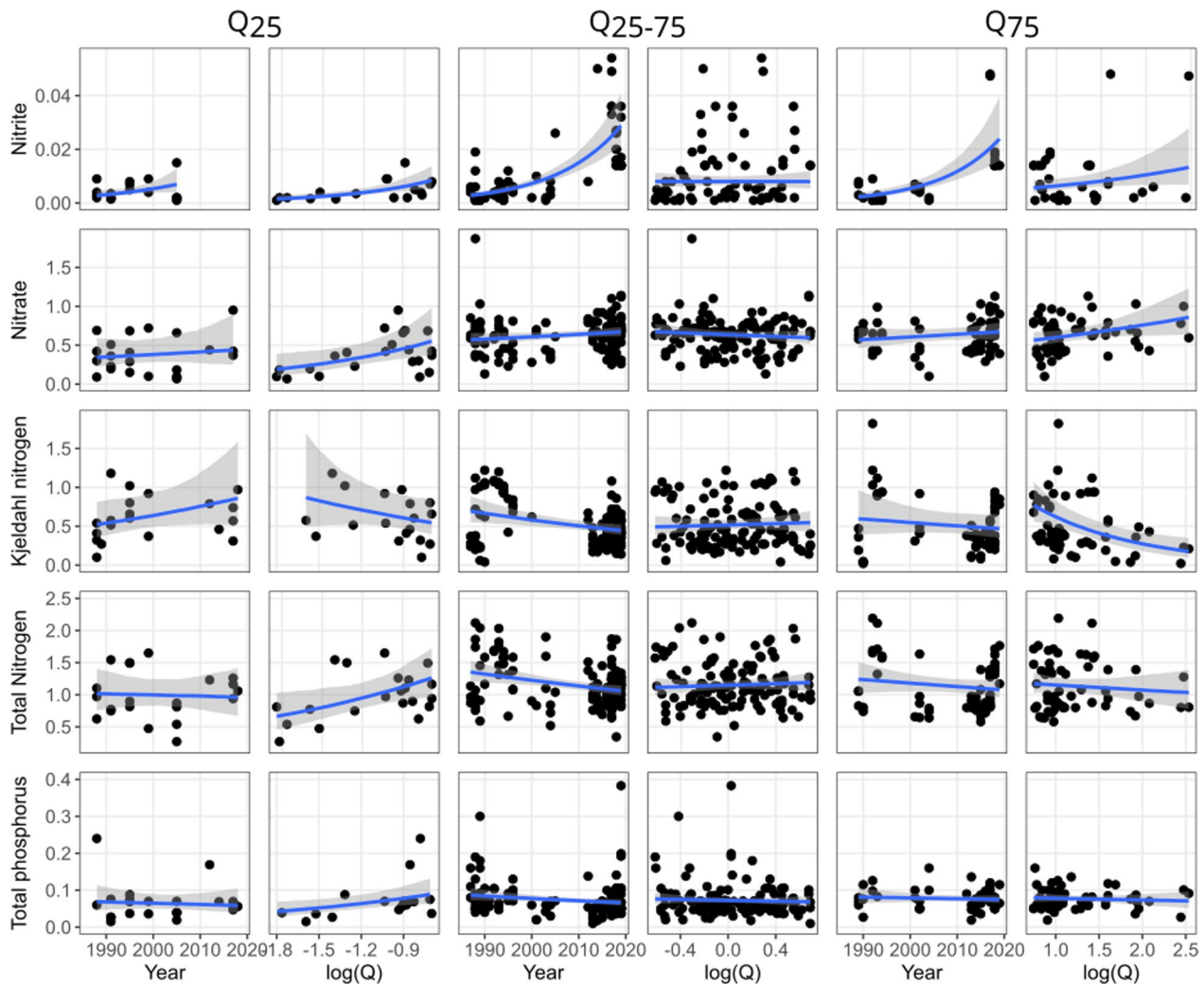


Fig. 6 Conditional predictive curves of Bayesian GLMs for nutrients as response variables as a function of year and log (Q) for each hydrological condition (Q_{25} : two columns of the left, Q_{25-75} : two columns of the middle, and Q_{75} : two columns of the right)

state (including nutrients, cyanobacterial abundance and chlorophyll-*a*) for this period was between Fair and Good (between 64.1 and 82.5; Fig. 8B). All nutrients and phytoplankton indicators contributed to the significance of the CCME-WQI calculation according to the Random Forest (Fig. S4B). The CCME-WQI of metals for the recent period was between Fair and Good (between 70.6 and 87.0; Fig. 8C). The variables that contributed to the significance of the CCME-WQI of metals according to the Random Forest were (in order of more to less important): zinc, chromium(VI), copper, iron, nickel and lead (Fig. S4C). The CCME-WQI for pesticides for the recent period (2018–2019) was between Good and Excellent (between 85.9 and 100.0; Fig. 8D). All the pesticides included in the CCME-WQI calculation contributed to the significance of the CCME-WQI calculation (Fig. S4D).

Finally, the overall CCME-WQI for the 2018–2019 period, calculated with all the variables that contributed significance in the partial CCME-WQI, showed Marginal–Good water quality (between 62.3 and 79.7; Fig. 8E). The Random Forest demonstrated that total phosphorus, chromium(VI), BOD₅, AOX, dissolved oxygen, zinc, atrazine and chlorpyrifos were the variables with the greatest importance in explaining the CCME-WQI (Fig. 8F).

Discussion

The study comprehensively evaluated temporal trends in water quality in the mid-lower Uruguay River, characterizing how specific water quality parameters have changed over the past 30 years. Discharge played a fundamental role in explaining variation in water quality parameters, particularly upstream the Salto Grande reservoir. Despite

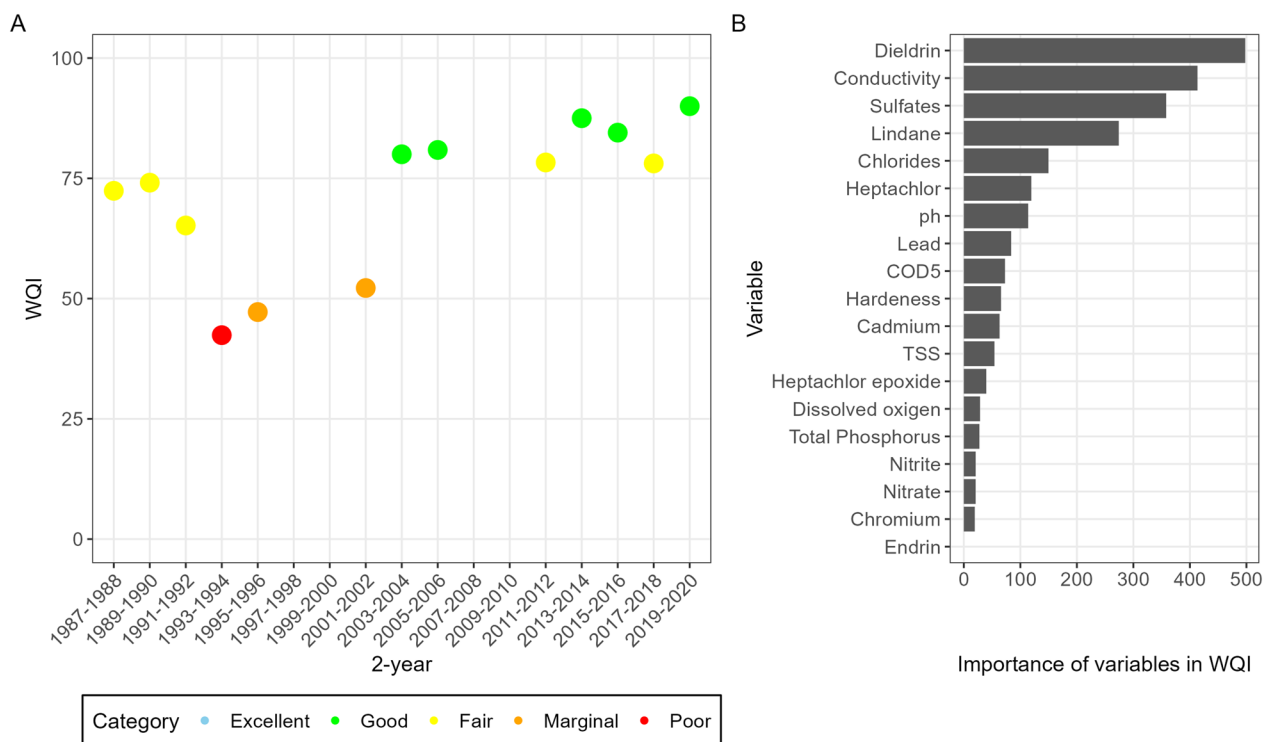


Fig. 7 **A** CCME-WQI in time (1987–2019). **B** Variable importance plot in the calculation of the overall CCME-WQI according to Random Forest (measured by total decrease in node impurities)

the increase in land use intensity, a deterioration of overall water quality was not observed. Rather, water quality in the mid-lower Uruguay River has improved over time for several parameters monitored along the whole period (e.g., present day-banned pesticides), resulting in increases in water quality index values along the last 30 years. However, during the analyzed period (1987–2019), water quality remained Poor to Good, and never Excellent. Several water quality parameters (e.g., metals and nutrients) were frequently non-compliant with respect to guideline values, with some concentrations increasing over time. In the recent period (2018–2019), modern monitoring including a higher diversity of parameters, also showed Marginal to Good status, but nutrients and phytoplankton indicators were associated with the lowest CCME-WQI values, indicating that eutrophication is in fact one of the main processes affecting water quality. In this section, we discuss the potential interpretations of our findings, focusing firstly on the historical variability in water quality, secondly the role of river discharge on water quality parameters, and finally implications for water quality management in the Uruguay River.

Historical change in water quality

One of the major concerns in the Uruguay River is the process of eutrophication, associated with increasing nutrient loads (Kruk et al. 2023). In this study, the reported total phosphorus (TP) values confirm that the mid-lower Uruguay River is a eutrophic river (average = 0.06 mg l⁻¹) (Chalar 2006; CARU Scientific Committee 2019a, b). The CCME-WQI of nutrients and phytoplankton indicators for the recent period showed that the high nutrient concentrations in the mid-lower Uruguay River are currently one of the main water quality problems. In a scenario of agricultural intensification associated with the replacement of natural grasslands by crops, especially since the 2000s (Modernel et al. 2016; Baeza and Paruelo 2020; Baeza et al. 2022), an increase in nutrients over time in the mid-lower Uruguay River was expected. However, total nutrients (TP and TN) and dissolved nutrients showed no relationship with time (except for nitrite that increased in concentration over time). The Uruguay River basin has historically been home to various industries and cities that have significantly impacted water quality (Teixeira De Mello et al. 2025). The impact of industries and domestic wastewater in large cities may be decreasing with the closure of certain industries and the implementation of industrial and domestic wastewater treatment plants (Alvareda et al. 2020; Teixeira De

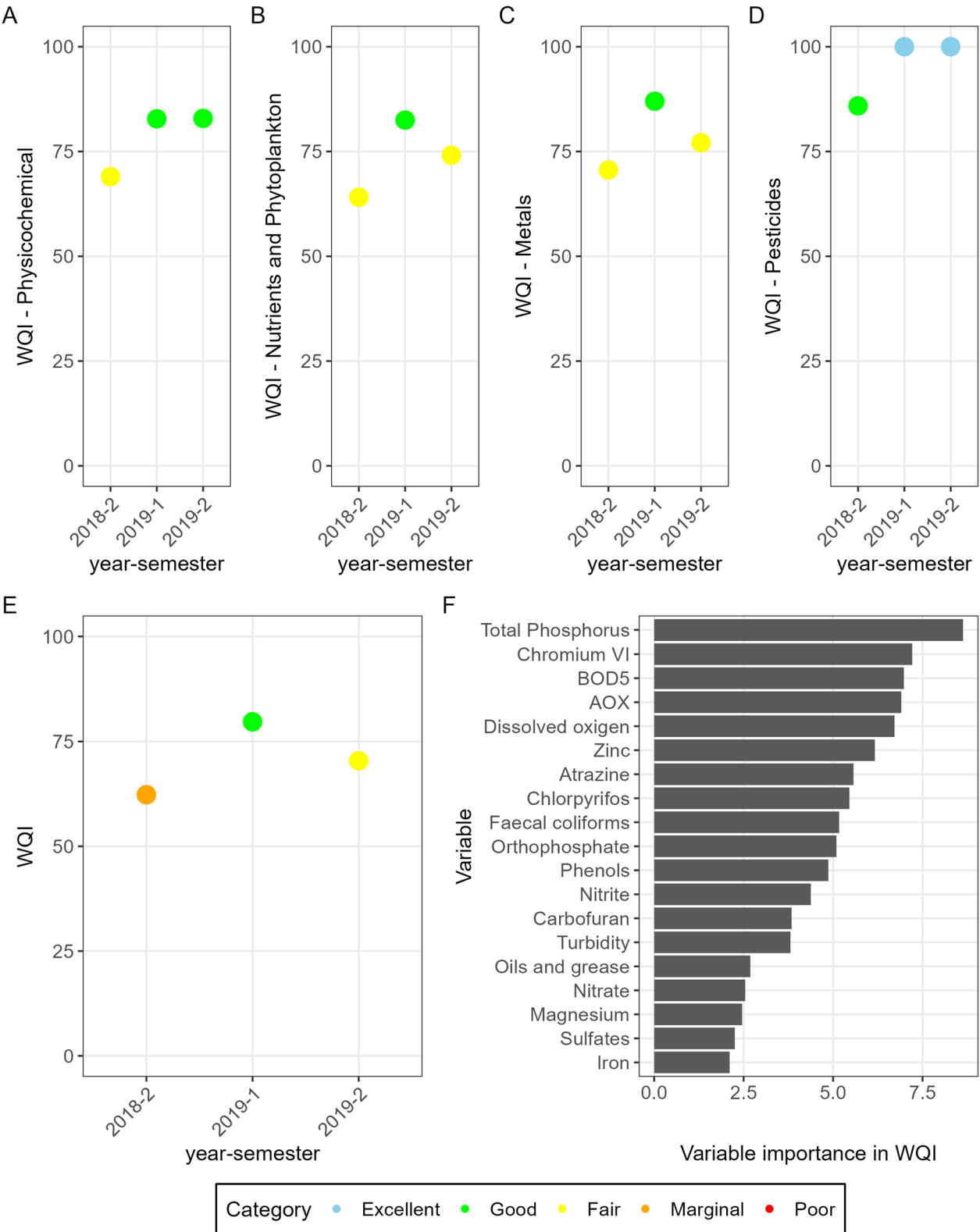


Fig. 8 Six-monthly CCME-WQI for the recent period (Jun/201–Dec/2019). **A** CCME-WQI for physicochemical parameters. **B** CCME-WQI for nutrients and phytoplankton indicators. **C** CCME-WQI for metals. **D** CCME-WQI for pesticides. **E** CCME-WQI for all types of variables. **F** Variable importance plot (measured by total decrease in node impurities) in the calculation of the overall CCME-WQI according to Random Forest

Mello et al. 2025). For example, several cities in Uruguay and Brazil have improved wastewater treatment plants (UN Water 2023; MVOTMA et al. 2020), which is in line with the decrease in fecal coliforms over time found by Kruk et al. (2023). Based on our results, we hypothesize that nutrient levels did not change over time due to the substitution of anthropogenic pressures, potentially shifting from nutrients originating predominantly in industrial and sewage point pollution sources to predominantly diffuse pollution sources from increasing agricultural inputs.

Increasing nutrients have been the proposed driver to an increasing frequency and magnitude of harmful algal blooms events in the last decades in the lower Uruguay River (Kruk et al. 2023). Although a trend of increasing nutrients was not supported here (and nutrient concentrations corresponding to eutrophic levels were found throughout the study period), other potential drivers for algal blooms have changed significantly over time. For example, a decrease in TSS and turbidity over time was found at RUY480, located downstream of 1000 km of an unregulated river stretch. This trend could be a consequence of the adoption of no-till farming as a technological improvement that reduces the input of organic matter derived from soil erosion (Cruz et al. 2011; Carrasco-Letelier and Beretta-Blanco 2017). Reduced turbidity may ultimately increase water transparency and light penetration that may promote the development of harmful algal blooms. Phytoplankton indicators, such as chlorophyll-*a* or cyanobacterial abundance, were not available in the 30-year database, hampering their analysis in this study. However, cyanobacterial blooms in lower Uruguay River are well documented, especially those originating in the Salto Grande reservoir (O'Farrell et al. 2012; Bordet et al. 2017; Martínez de la Escalera et al. 2017; Kruk et al. 2021) and in the three large reservoirs in the Negro River, the largest tributary to the lower Uruguay River (González-Piana et al. 2011, 2017; Aubriot et al. 2020).

The intensification of land use has been accompanied by new agricultural and livestock practices that involve an increase in the use of chemical compounds (e.g., fertilizers, pesticides and veterinary drugs) to increase production and control pests and diseases (Van Boeckel et al. 2015; Pascual Aguilar et al. 2017). This study reported a decrease in the pesticides aldrin, DDT, endosulfan, dieldrin, endrin, heptachlor, heptachlor epoxide, and lindane over 30 years. It is evident that this decrease in pesticide concentration, most of them reaching concentrations below the detection limit, is due to their total ban in the basin since 2005 (Law N° 17732/2003 in Uruguay (IMPO 2003) and Law N°26011/2005 in Argentina (Argentina Government 2004)). This reaffirms the importance of coordinated public policies at regional level

for the management of transboundary water resources (Saguier et al. 2021). However, in this study, only 8 traditional pesticides were analyzed consistently from 1987 to 2019, representing less than 5% of those currently used in Uruguay. As of December 2025, Uruguay had 227 active ingredients in plant protection products with various uses (e.g., herbicides, insecticides, fungicides, acaricides, or adjuvants) (data available at: MGAP 2026). The constant turnover in pesticide compounds during the last decades is also evident in the sediment record of Salto Grande reservoir, where traditional compounds such as DDT were abundant until 2000, while novel pesticides such as simazine become present since 2015 (Bardelle et al. 2026).

Recent water quality monitoring of the Uruguay River reported concentrations of more than 40 different pesticide residues in water (CARU Scientific Committee 2019a, b; Williman et al. 2017; Storck et al. 2026), of which at least 11 were recorded at concentrations higher than the limits that generate chronic and/or acute toxicity effects in biota (crustaceans and fish) (CARU Scientific Committee 2019a, b, c, d). Furthermore, residues of more than 30 pesticides bioaccumulated of fish muscle samples were detected in the Negro and Uruguay Rivers (Ernst et al. 2018; Storck et al. 2026). Not least, 16 pharmaceuticals were detected in fish species of the Uruguay River (Rojo et al. 2019). Also, microplastics were recorded in freshwater fish species in the middle Uruguay River (Santos et al. 2020). The constant emergence of new chemical products used in the agro-industry and pharmaceuticals implies a constant adaptation of both monitoring techniques and regulations, which represents a great challenge particularly for transboundary basins.

Metals in the mid-lower Uruguay River were highest in the 2000s (cadmium, chromium, copper, lead and mercury). Heavy metal pollution in surface water bodies from the Uruguay River watershed may have derived from increasing agricultural and industrial activities along with urban expansion (Avigliano et al. 2015) as well as mining (Da Silva et al. 2019). Historically (more than 100 years ago), mineral exploitation was an important activity in the upper Uruguay River basin, and actually the greatest extraction of amethyst geodes of the world (currently with 500 active and inactive mines) is in the municipality of Ametista do Sul (upper Uruguay River basin), which generates high concentrations of metals such as magnesium, copper, zinc, zirconium, barium, cobalt and cadmium that can be transported by rainwater to the rivers (Da Silva et al. 2019). Furthermore, several metals such as zinc and copper are also used as pesticides or are integrated in pesticide compounds (PPDB 2026). High concentrations of iron were frequently found throughout in Uruguay River associated with iron-rich

soils in the basin (Didone et al. 2014; Avigliano and Schenone 2016; Bardelle et al. 2026). High concentrations of lead above recommended levels found here during the 1990s and 2000s, were also reported for the upper Uruguay River (Avigliano and Schenone 2016) and in the Río de la Plata (Avigliano et al. 2015). Lead origin is attributed to effluents from large cities or industries located in the basin (Avigliano et al. 2015; Avigliano and Schenone 2016). Heavy metals are not naturally degraded over time, remaining in water, sediments or bioaccumulated in living organisms like fish (Paschoalini and Bazzoli 2021), ultimately generating a risk for these organisms and human health. For instance, mercury biomagnification in food webs of Uruguay River have been found in large specimens of piscivorous fish such as dorado (*Salminus brasiliensis*) with concentrations in fish muscle over the recommended by the Codex Alimentarius of FAO (Carrquiriborde 2018). In this context, the need to continue monitoring and enhance understanding on the dynamics and potential origins of metals in water, sediments and biota, will be important for future management.

Effects of hydrology on water quality parameters

The variability of water quality parameters in the Uruguay River has been associated with discharge patterns (Colombo et al. 2015). Consistently, this study found that turbidity increased with discharge, which could be an effect of the transport of particulate matter via runoff from the catchment. Nonetheless, the relationship between water quality and discharge was less evident in sites of the lower Uruguay River, downstream Salto Grande Dam. Several water quality parameters responded to changes in discharge at RUY480, including some physicochemical parameters (FSS, TDS, TSS) and nutrients (nitrate, total nitrogen and total phosphorus) increased with discharge, imply predominantly diffuse origin (Bowes et al. 2008). Particularly, while at low-water (Q_{25}), the positive relationship between nutrients and discharge remain significant, the lack of relationships at higher discharge scenarios may suggest that nutrient response to discharge rapidly reaches an asymptote, as suggested by nutrient-discharge models (Bowes et al. 2008; Goyenola et al. 2015). Meanwhile, the physicochemical parameters chlorides, conductivity, pH, sodium, and magnesium, and total hydrocarbons decreased with discharge, suggesting a predominantly point-source origin of contaminants (Bowes et al. 2008). This suggests that while nutrients and solids mainly originated from runoff in this largely agricultural basin, the ions mentioned, which affect conductivity and pH, and total hydrocarbons, may mainly originate in point sources such as industries and cities.

The physical barriers created by reservoirs affect the flow of nutrients and sediments downstream (Lehner et al. 2011). Reservoirs can act as “sinks” that retain nutrients and sediments (Tranvik et al. 2009; Némery et al. 2016; Liu et al. 2019), which could explain the site-specific differences in the response of water quality parameters to discharge up- vs. downstream the Salto Grande reservoir. However, the relationship between discharge and water quality at site RUY40 close to the Uruguay River mouth into the Río de la Plata estuary, may weaken due to the frequent occurrence of inverse flow, increasing the hydrological complexity of this zone (CARU Scientific Committee 2019a, b, c, d), which is influenced by the level of the Río de la Plata estuary, Paraná River discharge, wind direction and astronomical tides. These results reflect the hydrological complexity of large fragmented rivers and underscores the need for frequent water quality monitoring that integrates discharge measurements. Although, modern monitoring in Uruguay River includes hydrological measurements (CARU Scientific Committee 2019a, b, c, d), along this study discharge was extrapolating from locations with long term available at varying distances from the water quality monitoring sites.

Contributions to management in transboundary basins

CCME-WQI was a useful tool for assessing the overall water quality of the mid-lower Uruguay River. However, the use of WQI alone can lead to an underestimation of the effect of some variables as it reduces a large number of parameters to a single value (Gupta and Gupta 2021). Therefore, it is considered a good practice to accompany the use of WQI with an analysis of regulatory compliance for each parameter. In addition, the WQI disaggregated by Variable Type aided in understanding which type of pollutants were most relevant for the current water quality. This study demonstrates how the indices are sensitive to the parameters used in their calculation (Uddin et al. 2021; Fortes et al. 2023). Calculating the importance of each parameter in a WQI for each Variable Type using the robust statistical technique of the Random Forest model resulted a good approximation for selecting the subset of variables to include in the overall WQI (Lumb et al. 2011). In particular, pollutants with large fluctuations over a short time period, such as the numerous pesticides currently analyzed, could have biased the WQI result (Gupta and Gupta 2021). Further exploration of tools that comprehensively represent river water quality and that are directly applicable to establishing and monitoring management objectives and strategies are necessary (Hillebrand et al. 2025). The high variability of parameters used over time in long-term monitoring programs such as that explored here, and the inclusion

of emergent contaminants in historical comparisons, remain a critical challenge when using this tool as we experienced.

This study highlights the importance of long-term databases for quantifying changes in water quality and evaluating the impact of anthropogenic activities. It emphasizes the importance and challenges of maintaining long-term monitoring, as well as the challenges of synthesizing long-term data with non-consistent measurement frequency, changes in variables measured over time, and changes in sites monitored. Few biological variables were analyzed in this study because they were primarily included in the most recent period. The study also underscores the importance of maintaining standardized monitoring and systematizing water quality information generated by research groups and management institutions to enable water quality predictions and the development of more robust watershed management measures (Read et al. 2017).

Conclusions

This study is a comprehensive analysis of changes in 42 water quality parameters in the mid-lower Uruguay River over 30 years. We found that the concentration of some water quality parameters decreased (e.g., traditional pesticides), others increased (e.g., nitrite and chromium), and others remained stable over time (e.g., total phosphorus). In particular, we highlight the reduction of traditional pesticides, associated with the ban on their use by Uruguay, Brazil and Argentina. However, the constant changes in agricultural technologies and in the diversity of pesticides generate new emerging pollutants that have not been traditionally monitored and have a less clear role in water quality. WQI was a useful tool for integrating multiple parameters to evaluate overall river health and status, but understanding the behavior of individual parameters was an important aspect of its interpretation and future applicability to river management.

Discharge impacted temporal variation of certain water quality parameters, particularly upstream of the Salto Grande Dam. Moreover, the use of hydrological scenarios suggested differential effects of discharge on nutrient concentrations. This study points to the dominant role of diffuse source origins and highlights the influence of catchment runoff. This implies that land use management practices such as reducing fertilizer outputs from fields preventing erosion and nutrients losses, and the protection and enhancement of riparian buffer zones could be an effective strategy for improving water quality.

Finally, we emphasize the importance of long-term databases in large river ecosystems. Globally, these databases are relatively limited, and maintaining standardized monitoring over multiple decades is a major challenge.

Nonetheless, water quality databases and analytical tools for their analysis provide key inputs for water quality management (e.g., Jarvie et al. 2025). The database used in this study exhibited an information gap between 2005 and 2012, which coincides with the economic, political, and diplomatic crisis resulting from a pulp mill conflict between Uruguay and Argentina (Petherick 2010). This complexity of political, social, economic, and scientific pressures in development of the monitoring programs for a trinational river represents an additional challenge in the management of the Uruguay River. Along with this, the changing of pollutants used over time and the emergence of new pollutants create new challenges for monitoring programs over future decades.

Abbreviations

AMPA	Aminomethylphosphonic acid
AOX	Adsorbable organic halogen compounds
BOD ₅	Biological oxygen demand
CCME-WQI	Canadian Council of Ministers of Environment Water Quality Index
CARU	Uruguay River Management Commission
COD ₅	Chemical oxygen demand
DL	Detection limit
DDT	Dichlorodiphenyltrichloroethane
FSS	Fixed suspended solids
GLMs	Generalized linear models
K-W	Kruskal–Wallis test
Max	Maximum
Min	Minimum
MIRU	Comprehensive monitoring plan for the Uruguay River for the shared section
N	Amount of data different from NA
NA	Missing data
PCBs	Polychlorinated biphenyls
PIMI	Initial monitoring plan for the Uruguay River
PROCON	CARU's Water Quality and Contamination Control Program
Q	Daily discharge
Q ₂₅	Discharge below Percentile 25
Q _{25–75}	Discharge between percentile 25 and Percentile 75
Q ₇₅	Discharge above Percentile 75
RUY480	Monte Caseros-Bella Unión sampling site (30°13′S 57°37′W)
RUY90	Fray Bentos-Guaguaychú sampling site (33°07′S 58°22′W)
RUY40	La Concordia sampling site (33°34′S 58°28′W)
SNIIH	National water information system from Argentina
TDS	Total dissolved solids
TN	Total nitrogen
TP	Total phosphorus
TSS	Total suspended solids

Supplementary Information

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Supplementary Material 1.

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Author contributions

AS, IGB and CL participated in the conception and design of the study. AS and IGB made the data cleaning. AS analyzed the data. AS, IGB and CL lead the writing of the manuscript. All authors provided final approval for publication.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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